Adversarial Self-Auditing in an Autonomous Theory-Discovery Engine: A Case Study in Machine-Verified Cosmology

The OpenQG/ZYAL Collaboration

***Abstract*—We present OpenQG/ZYAL, an autonomous theory-discovery engine for late-time cosmology built around a single design rule: *the language model proposes; a deterministic oracle disposes*. Candidate modified-cosmology theories are emitted by an ensemble of free-tier language models behind a multi-provider router, expanded from a strict, machine-validated schema, *truth-bound* so that every certified modification alters the background the forward model actually integrates, and scored by a veto-first, 100-point rubric whose evidence term is a covariance-aware, Occam-charged likelihood over real distance, growth, and distance-ladder data. The engine's distinguishing feature is an institutionalized adversarial loop: every scoring generation's champions are attacked by multi-agent audits and an external 12-review referee cycle, and every exploit found is converted into a permanent regression test within hours. Across four rubric generations and three live campaigns (4,900+ generations of population evolution, 1,800+ ledgered LLM calls), this cascade eliminated reward hacks worth up to $+77/100$ points; its single most consequential discovery was *metrological*: a $+0.755$ (8.4σ) bias in the engine's own fitting-formula prediction of the CMB acoustic scale ℓ_A, which the optimizer had been harvesting for $\sim$35 nats of spurious evidence by drifting h. With the bias anchor-calibrated and complexity priced, the apparent h–Ω_m "degeneracy valley" ($\Delta \ln \mathcal{Z} \approx +55$ to $+69$) closes to $\Delta \ln \mathcal{Z} \approx -32$: the valley was an artifact of the instrument, not a property of the sky. The surviving candidate class — certified suppressed-growth phenomenology aimed at the $f\sigma_8$/S_8 tensions — improves the raw likelihood in the physically expected direction but does not yet clear the Bayesian evidence bar at the admitted data volume, a verdict the engine now reports, and defends, on its own.**

***Index Terms*—automated scientific discovery, large language models, evolutionary computation, reward hacking, cosmology, modified gravity, model selection, reproducibility**

I. INTRODUCTION

THE prospect of language-model–driven scientific discovery has moved rapidly from speculation to working systems: program-search engines have produced new mathematical constructions [1], end-to-end "AI scientist" pipelines draft and review their own papers [2], and learned models now lead entire subfields [3]. Yet the central obstacle to trusting such systems in the physical sciences is not generative capability but *verification*: an optimizer pointed at a numerical score will exploit every ambiguity in that score, a failure mode documented across decades of evolutionary computation [4] and formalized in the reward-hacking literature [5], [6], [7].

This paper reports a system designed from the outset around that adversarial reality, and a multi-day, fully-ledgered case study of what it took — four successive rubric generations, three live model campaigns, two external referee cycles, and one 21-agent internal audit — before the engine's own headline results stopped being exploits. Our domain is late-time cosmology: the persistent tensions between early- and late-universe measurements of the Hubble constant H_0 [8], [9] and of structure growth S_8/$f\sigma_8$ [10], [11] define a live scientific frontier in which candidate modified-gravity and dark-sector theories [12], [13], [14] can be expressed compactly, scored against real data, and falsified mechanically.

The contributions are:

1) **An architecture for honest LLM-driven theory search** (Sec. III): strict-schema proposal sketches expanded deterministically into typed theories; *truth-binding*, which compiles verified derivation certificates into the background cosmology the forward model integrates, so that claims have computable consequences; and a veto-first scorecard in which every credit dimension is machine-recomputable.
2) **An evidence economics for generative search** (Sec. IV): covariance-aware likelihoods over real survey data [15], [16], a BIC-style Occam term [17], [18] charged on *every* moved degree of freedom — including silently drifted background coordinates and proposer-chosen certificate inputs — and mechanism-attributable novelty, in which a "novel prediction" must survive comparison against a mechanism-off twin on the same background.
3) **The audit cascade as a first-class product** (Sec. VI): a documented sequence in which champions scoring 87.5, 55.0, 77.0, and 60.0 under successive rubrics were each disqualified by the next generation's audit, with every exploit converted into a permanent regression test; we argue this cadence, not any single score, is the correct success metric for autonomous discovery systems.
4) **A metrological result with scientific content** (Sec. VII): the engine's optimizer located an 8.4σ bias in the system's own CMB fitting formula and converted it into spurious cosmological evidence; after anchor calibration in the spirit of [19] and honest complexity pricing, the apparent open "degeneracy valley" toward high h closes decisively.
5) **An honest candidate verdict** (Sec. VIII): the surviving

Manuscript prepared June 2026. The complete system — engine source, streamed campaign ledgers, replay tooling, audit records, and external review archives — is versioned in the OpenQG repository (branch `chore/v6-purge`); every quantitative claim in this paper is reproducible from committed artifacts via `ops/replay.sh` and `zyal genome rescore`.

theory class — certified suppressed-growth phenomenology in the Planck μ_0 parametrization [20] backed by a dark-scattering mechanism relation [14], [21] — moves the growth observables toward the data but does not clear the evidence bar at current data volume. The engine reports this itself, and its report survives external review.

II. Related Work

LLM-driven discovery. FunSearch [1] pairs an LLM proposer with a deterministic evaluator over programs; our engine adopts the same proposer/oracle asymmetry but for physical theories, where the evaluator must itself be defended against exploitation — the oracle is not a fixed ground-truth function but a scientific instrument with its own biases (Sec. VII). The AI Scientist [2] automates the paper-writing loop with LLM reviewers; we instead route review through a deterministic rubric plus an external LLM referee whose findings must be reproduced as failing tests before they count.

Evolutionary search. Population structure follows the quality-diversity tradition [22], [23]: explore/exploit/novelty islands over typed theory genomes, with a re-clothing operator (Sec. III) that grafts verified derivational structure onto evolved parameter sets — an explicit structure$\times$fit recombination. The exploit catalogue our cascade generated is a domain-specific instance of the anecdotes in [4].

Reward hacking. Our findings track the taxonomy of [6]: proxy gaps between "fits the admitted data" and "is good cosmology" were exploited through specification holes (uncosted drift), measurement holes (the ℓ_A bias), and verification holes (self-certified novelty). The engineering response — converting each discovered hack into a vetoed, regression-tested behavior — operationalizes the mitigation programme sketched in [5], [7].

Cosmological model selection. Scoring builds on standard information criteria [24], [17], [25], [18], compressed CMB distance priors [15], DESI DR1 BAO with published per-tracer covariances [16], RSD growth compilations [11], and the local distance ladder [9]. The modified-gravity vocabulary spans nDGP [12], [26], Hu–Sawicki $f(R)$ [13], the Planck μ_0 parametrization [20], and dark scattering [14], [21], gated by solar-system and multimessenger constraints [27], [28], [29], [30], [31].

III. System Architecture

Fig. 1 shows the V7 system. We describe the bands in proposal order.

A. Proposal: strict sketches over a model ensemble

Proposals are requested from a local *fusion router* that fronts 131 free-tier models across 16 providers with win-rate–based routing, quality-band selection, and per-call winner-model telemetry. Each proposal slot fires $K{=}4$ parallel samples, each assigned a deterministic *mechanism lane* — `planck_mu0` (suppressed-growth phenomenology), `dark_scattering` (the mechanism-backed lane), `free` (model's choice), and `null_diagnostic` (an adversarial control asked to produce the most ΛCDM-degenerate honest theory) — so that single-model monoculture cannot collapse the search.

Models do not emit the engine's internal serde types. They emit a flat *ProposalSketch* validated server-side against a strict JSON schema whose enumerations (registry relations, physics sectors, obligation kinds) are generated from the engine's own source of truth and locked by anti-drift tests. A pure, deterministic expander turns sketches into typed theories; in V7 the expander also attaches the *generating term* for every certified relation and turned dial (a brane term for nDGP, a coupling term for dark scattering), so structural completeness is preserved by construction for honest proposals. The yield consequence was decisive: the V5 campaign lost 77% of calls to transport failure and $\sim$90% of survivors to parse errors; under the schema'd router, parse failures fell below 4% and transport empty-replies to zero (Fig. 2).

Failed proposals are not discarded: a parse failure triggers one repair round carrying the exact expander error; an oracle kill triggers one repair round carrying the kill reasons — with engine-computed witness values *redacted* in V7, so the model is taught the contract, not the oracle's numerical surface. Every call, repair, and failure is a ledger record with its winner-model identity, lane, quality band, and content hash.

B. Truth-binding: claims with computable consequences

The single most important integrity mechanism is *truth-binding*. A proposal that certifies a modification — e.g. $G_{\mathrm{eff}}/G = 1 + 1/(3\beta)$ via the nDGP relation with $\beta = 2$ [26] — has that certificate *compiled into the background the forward model integrates*: the binder inverts the certified coupling into Ω_{rc} and sets the nDGP family on the integrated cosmology. A certified claim therefore has computable consequences for every observable; a claim the engine cannot implement, cannot explain, or that conflicts with the declared background is a kill (`Unimplemented/Unexplained/ConflictingModification`). In V6.1 this was extended to *claim/physics coherence*: an obligation certificate whose inputs disagree with the parameter certificate the binder used (rigor earned at $\mu_0{=}-0.05$ over a fit run at $\mu_0{=}-0.1$) is likewise a kill.

C. The unified physics gate

No candidate is scored while `physics_kills` returns a reason. The gate fuses the triage cascade (dimensional homogeneity, ghost exclusion, certificate verification, evidence binding) with *adjudication*: quantified screening — a bare mechanism label such as "Vainshtein" is insufficient; the candidate must carry a numeric recovery fraction whose residual passes the Cassini bound [27] — the tensor-speed constraint evaluated at the GW170817 source redshift [30], [31], background physicality at every probe redshift, phantom-drag exclusion, and, in V7, the *term registry*: every turned dial must name an explicit generating term (`dgp_brane`, `f_r_correction`, `quintessence_scalar`, `dark_scattering_coupling`, ...), including dials reached *through binding* — the check that retroactively disqualified the V6 campaign's only surviving champion (Sec. VI).

placeholder

Fig. 1. The OpenQG/ZYAL V7 engine. *Proposal band:* a local fusion router fronts 131 free-tier models across 16 providers; the RouterProposer fires best-of-K=4 samples per slot, each pinned to a mechanism lane, with server-side strict-JSON-schema validation and redacted parse/oracle repair loops. *Oracle band:* sketches expand deterministically into typed theories (generating terms travel with every turned dial); truth-binding compiles verified certificates into the integrated background; the unified physics gate (triage cascade + adjudication) applies the term registry, quantified screening, the GW170817 tensor-speed bound at source redshift, and phantom-drag exclusion before any scoring; the 100-point scorecard and the Occam-charged covariance evidence term complete the verdict. *Evolution band:* island-structured population search with the re-clothing (structure×fit) operator and promotable-gated cross-run memory. *Audit band:* streamed attempt/proposal ledgers, deterministic replay, and the adversarial audit loop whose findings return to the oracle as permanent regression tests.

D. Evolution and the re-clothing operator

Surviving candidates breed in an island-structured population (explore/exploit/novelty) in the quality-diversity tradition [22], [23], with deterministic seeded RNG so that whole campaigns replay exactly from content-pinned ledgers. The V6 ceiling analysis showed proposals carrying derivational *structure* without optimized *fit*, while evolved children carried fit without structure. The *re-clothing* operator closes the gap: every 25 generations the best live proposal's claim graph (and its certified claim-carrier parameters and generating terms) is grafted onto the best evolved theory, novel-prediction witnesses are *refreshed to the engine-computed truth on the descendant's bound background* and flagged `refreshed_by_engine`, and the graft is scored through the full oracle with nothing copied from the donor scorecard. Grafts onto drifted descendants forfeit the donor's unification claim. Every V6 and V7 campaign champion is a re-clothed lineage.

IV. Scoring and Evidence Economics

Table I summarizes the rubric. Three V6.1/V7 mechanisms deserve emphasis because each closed a live exploit.

1) Complexity is charged everywhere: V5's champions moved h, Ω_m, and w_0 "for free": parsimony counted only declared parameters. V6 introduced `background_dof` — any standard coordinate moved off the Planck reference is a fitted dial unless a verified relation derives that exact value (and a bare "fundamental" declaration never exempts). V7 extended the ledger to *post-search choices*: a proposer-chosen mechanism input wearing a verified certificate (β=2, $A_{\rm drag}$=2, μ_0=-0.1) is still a chosen dial and costs a degree of freedom [18]. The same k feeds the Occam term, with $n_{\rm eff}$ counted as effective modes (per-block member counts plus unblocked records) rather than raw record count.

2) The likelihood is covariance-aware and mode-stamped: All scoring flows through one likelihood path: correlated survey blocks — the Planck distance-prior 3×3 over $(R, \ell_A, \omega_b h^2)$ [15] and DESI DR1 per-tracer $(D_M/r_d, D_H/r_d)$ blocks [16] — enter as covariance blocks with positive-definiteness checks and marginalization over missing members; everything else is an independent Gaussian. Every `DataFitOutcome` carries its `likelihood_mode` and block count, so no headline number can hide its statistical treatment. Notably, the move from diagonal to true covariance *weakened* the compressed CMB constraint along the drift direction (Sec. VII): the diagonal treatment had been double-counting two strongly correlated numbers [15] as independent.

3) Novelty is mechanism-attributable: The V6 campaign's +17-point champion earned full novelty from an honest, computable witness on ℓ_A — an observable *in the fit set*, whose deviation was produced by the background drift rather than the certified mechanism, and whose declared value had been authored by the engine's own re-clothe refresh. V7 novelty requires all three repairs at once: distinctness is measured against a *mechanism-off twin* (the bound background with every MG dial returned to its GR limit, so drift cancels), fit-set witnesses cap at quarter credit, and engine-refreshed witnesses are flagged on the audit record and capped at half credit.

TABLE I
THE V7 SCORECARD. EVERY DIMENSION IS MACHINE-RECOMPUTABLE; "DEMOTE" ROWS ZERO THE DIMENSION AND FLAG THE AUDIT RECORD; "KILL" ROWS DISQUALIFY.

Dimension (pts)	**Machine semantics**
Derivation rigor (20)	Obligations re-verified against the registry of closed-form relations; rigor weighted per relation (a cited parametrization earns 0.3; a mechanism with integrated equations 0.8–1.0). Certificate/binding incoherence kills.
Data fit (20)	$\Delta \ln \mathcal{Z} = \Delta\mathrm{LL} - \frac{1}{2} k_{\mathrm{eff}} \ln n_{\mathrm{eff}}$ against profile-anchored ΛCDM under the covariance likelihood; ΔAIC carries $2k$ [24], [17]. Mode and block count stamped on every scorecard.
Novel prediction (20)	Witness must be *computable, honest* ($\leq$ engine-clamped tolerance; $> 3\times$ is a fabrication kill), and *mechanism-distinct*: compared against a mechanism-off twin on the same background, so drift-bought shifts cancel. Fit-set observables cap at 0.25; engine-refreshed witnesses at 0.5.
Unification (15)	Shared parameters must exist, drive ≥ 2 sectors, agree with the integrated background (shadow-consistency kill), and hide no knob: drifted background coordinates count as hidden knobs.
Robustness (13)	Sealed alternating-holdout generalization gap (unclamped).
Parsimony (12)	Linear $1 - k/4$ over k = free parameters + drifted background coordinates + proposer-chosen certificate inputs (post-search choices).

TABLE II
THE ADMITTED MULTISECTOR DATASET (V6.1/V7 EVIDENCE PATH). COMPRESSED CMB AND PER-TRACER BAO SECTORS CARRY PUBLISHED COVARIANCE BLOCKS; THE REMAINDER ARE INDEPENDENT GAUSSIANS. "ΛCDM PULL" IS $(x_{\mathrm{obs}} - x_{\Lambda\mathrm{CDM}})/\sigma$ AT THE PLANCK-ANCHORED BASELINE.

Sector	**Records**	**Covariance**	**Character**
H_0 ladder [9]	1	diagonal	$+5.4\sigma$ pull
S_8 lensing [10]	1	diagonal	-3.2σ pull
RSD $f\sigma_8(z)$ [11]	5	diagonal	uniformly low
BAO $D_{M,H,V}/r_d$ [16]	12	5 tracer blocks	anchor
CMB $(R, \ell_A, \omega_b h^2)$ [15]	3	3×3 block	anchor (calibrated)
BBN Y_p	1	diagonal	anchor

V. LIVE CAMPAIGN INFRASTRUCTURE AND BEHAVIOR

Each campaign runs as six 300-generation chunks (population 12) on a pinned engine binary, with proposal slots every 20–40 generations, streamed ledgers, and atomic champion checkpoints; chunked execution made campaigns kill-resistant on a heavily loaded host, and cross-run memory (promotable-gated in V6.1: only candidates that pass the *current* unified gate may become exemplars) compounds learning across chunks. Fig. 2 shows the funnel evolution; Fig. 3 the full activity rasters. Two behavioral observations matter for system designers. First, *the oracle is the teacher*: oracle-repair rounds carrying kill reasons converted a substantial fraction of killed first attempts into scored proposals, and the dominant residual kill class shifted from physics vetoes to certificate-plumbing friction — addressed in V7 by publishing the exact per-relation input keys and a verbatim-passing certificate example in the lane prompts — after which every lane's OK yield rose (planck_mu0 18 → 31, dark_scattering 5 → 9, free 10 → 13, null_diagnostic 19 → 20; Fig. 2, bottom). Second, *routing concentrates*: win-rate routing sent a majority of OK calls to a single 120B-class open-weight model (with a 20B sibling and one 120B competitor covering most of the remainder); the ledgered winner-model telemetry makes this measurable, and lane assignment keeps the search mechanism-diverse even when the model pool is not.

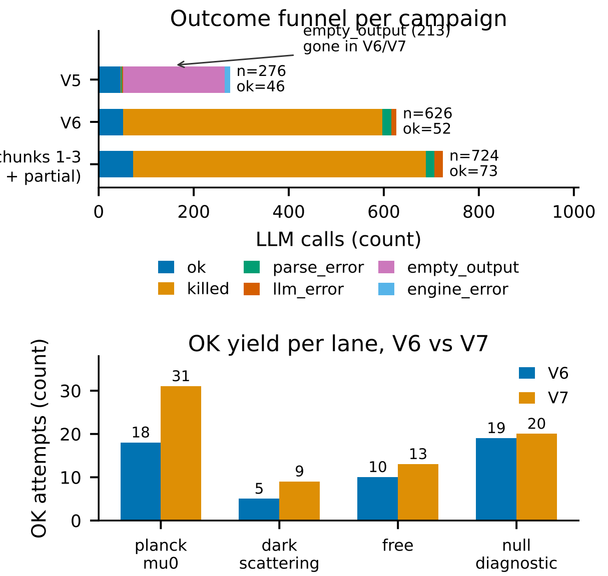

Fig. 2. Top: attempt funnels for the three campaigns. The V5 subprocess transport lost 213/276 calls to empty replies; the schema'd HTTP router eliminated the class entirely, after which the dominant outcome is *killed* — the oracle, not the transport, became the binding constraint. Bottom: per-lane OK yields under V6 and V7; the adversarial `null_diagnostic` control converts most often, and the mechanism-backed dark-scattering lane improves after the V7 per-relation input-key contract.

TABLE III
THE CASCADE: EACH RUBRIC GENERATION'S CHAMPION, THE EXPLOIT THAT PRICED IT, AND THE FIX THAT KILLED IT. EVERY ROW IS NOW A COMMITTED REGRESSION TEST.

Champion	**Score**	**Exploit → fix**
V4 fixture era	87.5	rediscovery credit, tie-credit, structural unification → V4.1 integrity rubric
V5 campaign (×6 seeds)	55.0	free background drift; bare screening labels; α-jitter half-novelty → V6 unified gate, drift cost
V6 chunk 4	77.0	fit-set ℓ_A witness, self-certified by re-clothe refresh, riding the ℓ_A instrument bias → V6.1 calibration, mechanism-off novelty, coherence kill
V6 chunk 6 survivor	60 → 25	bound nDGP with no brane term; costless certified β → V7 term registry, post-search-choice dof
V7 campaign	43.0	*standing*

VI. THE AUDIT CASCADE

Fig. 4 and Table III document the central result of this case study as a process: *four consecutive rubric generations*

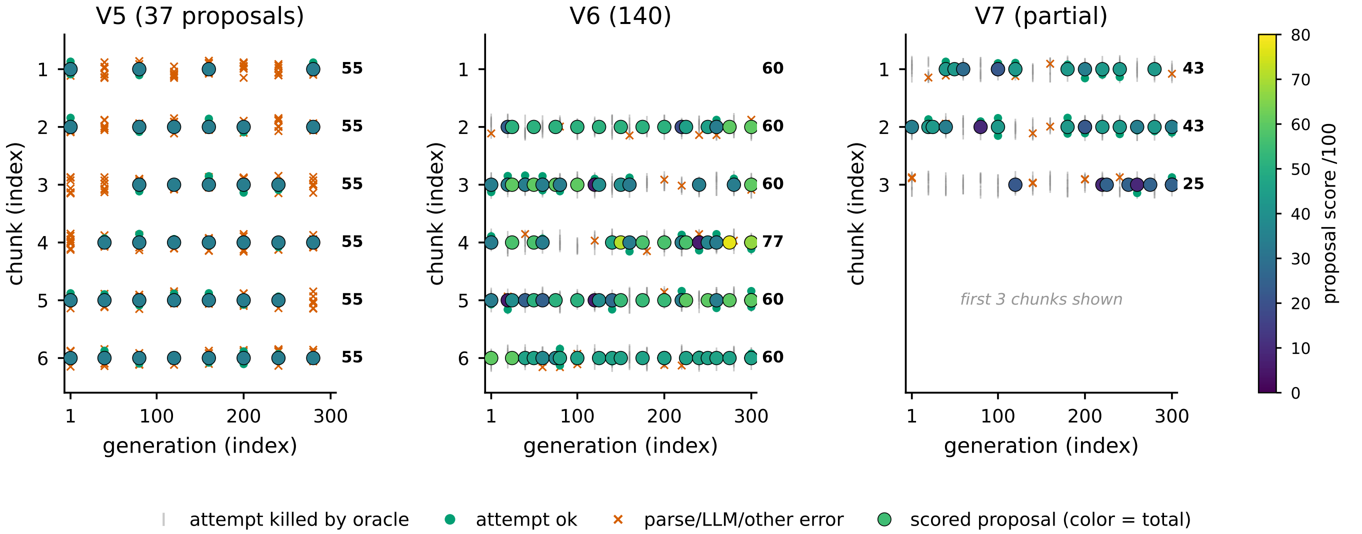

Fig. 3. Campaign activity rasters: every ledgered LLM attempt (kills, errors, OK calls) and every scored proposal (colored by score) across generations and chunks, for the V5, V6, and V7 (first three chunks) campaigns. Scored-proposal density rises era over era (37 $\rightarrow$ 140 $\rightarrow$ 160^{+}) even as the oracle hardens, because repair loops, compounding cross-run memory, and the strict-schema transport convert more of the model ensemble's output into adjudicable physics.

in which the engine's own champions were adversarially invalidated, each within hours of the audit that exposed them. Three properties make this cascade more than iterative bug-fixing. First, audits are themselves multi-agent and adversarial: the V6.1 round ran 21 parallel reviewer/refuter agents over code and ledgers, with every high-severity finding surviving an explicit refutation attempt before being actioned. Second, external referees are in the loop: two 12-review cycles by an independent frontier model (prompted with the full source, ledgers, and the team's own self-criticism) produced the go/no-go verdicts and several findings the internal audits missed — including the survivor's missing brane term. Third, and decisive for trust: *every* finding lands as a vendored, permanently failing-then-fixed regression test, and prior champions are mechanically re-scored under each new rubric (`zyal genome rescore`), so the cascade's history is replayable, not anecdotal.

Table V summarizes the V6.1 internal audit that anchored the cascade's sharpest turn. The findings cluster into recurring classes worth naming for future system designers: *instrument* faults (the evaluator's own physics is wrong); *laundering* (credit earned on one object collected on another — grafted unification scaffolds, engine-authored "honesty"); *specification* holes (a deferred cap or an over-broad exemption becomes load-bearing the moment an optimizer touches it); *coherence* gaps (two internally-consistent subsystems disagreeing about the same quantity); *economics* (any unpriced degree of freedom is free energy); and mere *friction* (schema misspellings), which is the only class repair prompting should solve — the others must be vetoes. The empirical regularity across all four generations: the optimizer found the cheapest open class first, and exhausted each class before moving to subtler ones. A deferred integrity feature is therefore not a roadmap item; it is the next champion.

VII. The Degeneracy-Valley Resolution

The scientific climax of the case study is a result the engine produced *about its own instrument*, in three acts (Fig. 5).

Act I (V5): the valley opens. Six independent campaign seeds converged on the same champion: an evolved background at $h \approx 0.718$, $\Omega_m \approx 0.282$, $w_0 \approx -1.14$, holding $\Omega_m h^2$ nearly fixed — the classic compressed-CMB degeneracy direction — and harvesting $\Delta \ln \mathcal{Z} \approx +54.9$ by relieving the H_0 and S_8 pulls simultaneously. The convergence was genuine; the interpretation was not.

Act II (V6): covariance makes it worse. Replacing the diagonal likelihood with the published distance-prior covariance [15] *raised* the credit to $+68.8$ nats: along the degeneracy direction, correlated residuals are cheaper than independent ones, so honest statistics widened the apparent valley. This is itself a cautionary result for compressed-likelihood analyses.

Act III (V6.1): the instrument confesses. A 21-agent adversarial audit traced the evidence budget and found that $\sim$35 of those nats were flowing from a $+0.755$ (8.4σ under the prior's $\sigma_{\ell_A} = 0.090$) bias in the engine's *own* fitting-formula prediction of ℓ_A at the Planck anchor — fitting-formula physics meeting Boltzmann-grade data precision [32]. The optimizer had discovered the instrument's flaw and weaponized it through $\partial \ell_A / \partial h$. Anchor calibration (precedented by the BAO sound-horizon treatment of [19]), a regression guard pinning the ΛCDM baseline to the published priors, and the Occam charge on the drift's three degrees of freedom flip the verdict to $\Delta \ln \mathcal{Z} = -31.9$ under covariance: **the valley was never open**. We regard this as the strongest single argument for adversarial self-auditing in autonomous science: a generative optimizer is an exquisitely sensitive detector of its own evaluator's biases, *if* the system is instrumented to trace where its winnings come from.

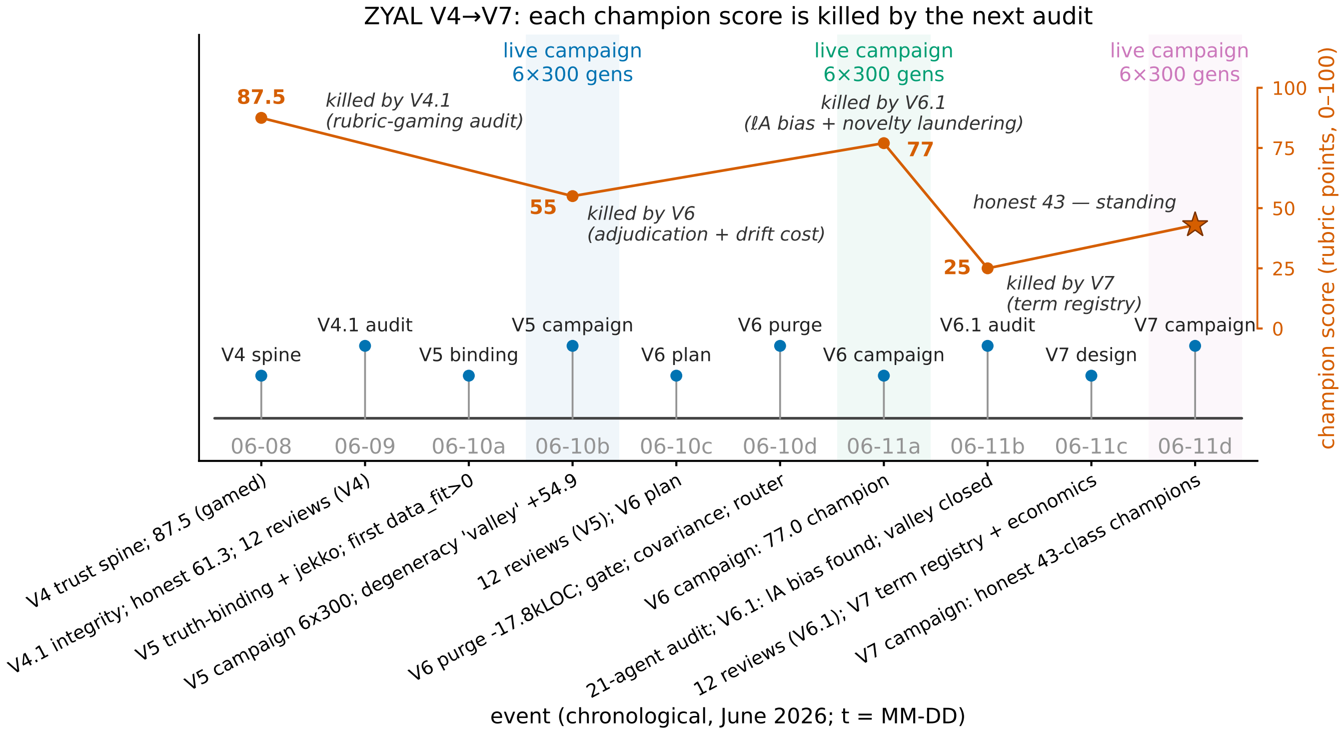

Fig. 4. The ZYAL arc: ten build/audit/campaign events across four rubric generations, overlaid with the champion-score trajectory. Every peak is killed by the following audit: V4's 87.5 (rubric gaming) by the V4.1 integrity pass; V5's 55.0 class (uncosted drift, unquantified screening) by the V6 unified gate; V6's 77.0 (self-certified fit-set novelty riding an instrument bias) by the V6.1 calibration and novelty overhaul; the V6.1 survivor's 60→25 re-cost by the V7 term registry. The V7 campaign's 43-class champions are the first whose every point is re-earnable under the standing rubric.

TABLE IV
V7 CAMPAIGN CHUNK RESULTS TO DATE (CAMPAIGN IN FLIGHT AT SUBMISSION; THE STANDING RUBRIC). CHUNK-3'S CHAMPION IS AN HONEST NULL: A RE-CLOTHED ΛCDM LINEAGE CARRYING NO DISTINCT PHYSICS.

Chunk	Champion	Score	Distinct	Proposals
1	`g0040-proposer (planck_mu0)`	43.0	yes	27
2	`planck-mu0-suppressed-0.1`	43.0	yes	28
3	`gr-lcdm-baseline-rc`	25.0	no	24
4–6	*in flight on the pinned V7 binary*			

VIII. RESULTS: THE HONEST CANDIDATE CLASS

A. The V7 champion class

Under the standing rubric, the V7 campaign's champions to date score 43.0 (chunks 1–2, both `planck_mu0`-lane lineages; chunk 3's honest null was a re-clothed ΛCDM baseline at 25.0). The 43-class decomposition (Fig. 7) is, by construction, fully re-earnable: derivation credit at the parametrization ceiling (0.3 rigor weight — the engine refuses to pretend phenomenology is mechanism), unification from genuinely shared and background-consistent parameters, robustness from a clean sealed-holdout gap, parsimony charged for the chosen μ_0, and *zero* data-fit credit — because the mechanism's ∼0.5-nat raw improvement does not clear the $\frac{1}{2}k \ln n_{\rm eff} \approx 0.9$-nat evidence bar at the admitted data volume [17], [25].

B. The candidate physics, stated honestly

The surviving theory class is: *late-time growth suppression, expressed as the Planck $\mu(a) = 1+\mu_0\,\Omega_\Lambda(a)/\Omega_\Lambda$ parametrization with $\mu_0 < 0$ [20], with dark scattering — a DE–DM momentum-exchange drag $\Gamma(a) = A_{\rm drag}(1 + w(a))\,\Omega_{\rm de}(a)$ added to the linear growth equation [14], [21] — as the registry's first mechanism-backed route to it.* Fig. 6 shows why the ensemble converges here: the admitted growth data sit uniformly low, and suppression bends $f\sigma_8(z)$ toward every growth point at once. The engine's verdict, which we endorse and emphasize: *right direction, insufficient evidence at current data volume*. The candidate is falsifiable along two axes the system already encodes — more growth data tightens the raw likelihood gain against a fixed Occam bar, and the dark-scattering lane's $(1 + w)$ factor ties the mechanism to a measurable $w_0 > -1$ preference.

C. Comparison with the V6 contender league

Against the engine's human-program baselines (a GR+ΛCDM program and weaker textbook contenders, scored under the same oracle), the V7 champion class sits below the strong baseline — as it should: a phenomenological parametrization that cannot yet beat ΛCDM on evidence has no business outranking it. We consider this ordering itself a system result: four rubric generations ago, the ordering was inverted by exploits.

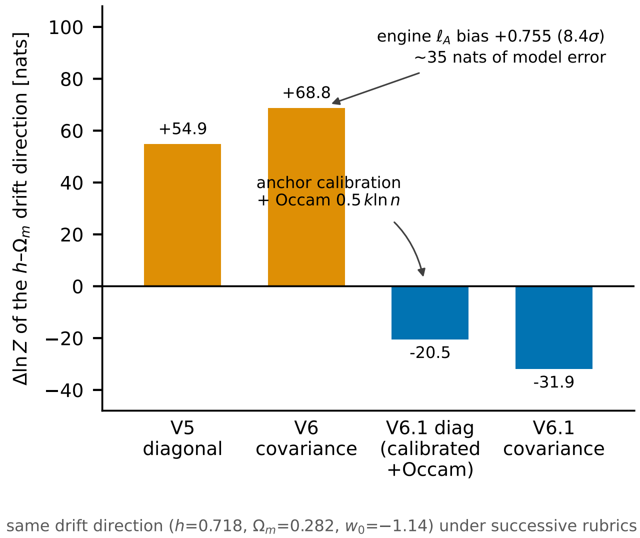

Fig. 5. Evidence assigned to the same h–Ω_m–w_0 drift direction under successive rubrics. Diagonal scoring awarded $+54.9$ nats; the true correlated distance-prior covariance *raised* it to $+68.8$ (the diagonal treatment had overstated the compressed CMB's constraining power); anchor-calibrating the engine's ℓ_A fitting formula and charging the Occam term closes the valley to -20.5 (diagonal) and -31.9 (covariance). The apparent discovery was an instrument artifact.

IX. Reproducibility and Artifact Contract

The engine treats reproducibility as a scoring-grade obligation rather than a packaging afterthought, because the audit cascade is only trustworthy if its history is mechanically re-derivable.

1) Determinism contract: Population evolution is seeded and pure: two runs of the same configuration are bit-identical, a property enforced by a trust-gate that re-runs the engine twice and compares champions before any campaign is allowed to launch. Live LLM calls are inherently nondeterministic, so campaigns never replay from seeds: they replay from *content*. Every proposal is ledgered with its full document and SHA-256; `zyal genome replay` re-scores a ledger with no network access and reports mismatches (zero across all campaigns reported here, including replays of V6-era ledgers under the V6-era pinned binary).

2) Total observability: Every LLM call — including every parse failure, repair round, transport error, and oracle kill — is an attempt record carrying outcome, kill reasons, raw-output hash and length, winner-model identity, lane, quality band, and upstream repair count. The funnels of Fig. 2 and the rasters of Fig. 3 are direct ledger aggregations, not reconstructions. Streaming sinks write records as they are produced with atomic champion checkpoints every 25 generations; two externally-killed campaign processes on a contended host lost zero records.

3) Re-scoring as the audit instrument: The `rescore` subcommand loads any archived champion and adjudicates it under the current rubric; every number in Table III is its output. Pinned campaign binaries keep in-flight datasets rubric-consistent while the development tree advances, and sealed replay expectations (re-issued as `v2` when the ℓ_A calibration legitimately moved league numbers, with `v1` retained as the pre-calibration record) let an external referee re-derive headline league numbers with one command.

4) Cost profile: All proposal-side inference ran on free-tier routed models; the only billable LLM usage was the external referee cycles. The deterministic oracle and campaigns ran on a single shared commodity host. We consider this profile — free generation, expensive verification deliberately rate-limited to referee checkpoints — the economically sustainable shape for autonomous-science loops.

TABLE V

The V6.1 internal audit at a glance: 14 adversarially-verified findings from a 21-agent workflow (4 reviewer dimensions × parallel refutation), all fixed and regression-locked within one day. Two further candidate findings were refuted by the verification stage and discarded.

Finding (abridged)	Class
Engine ℓ_A formula biased $+0.755$ (8.4σ); optimizer harvesting $\sim$35 nats	instrument
Re-clothe witness refresh manufactures "honest" novelty	laundering
Fit-set witness earns full novelty (deferred cap was load-bearing)	specification
Grafted unification scaffold contradicts fitted background	laundering
Obligation certs incoherent with binder's certs (μ_0=-0.05 vs -0.1)	coherence
Background-dof exemption accepted any verified certificate	specification
Negative drag = uncertified growth enhancer (sign asymmetry)	implementation
Drag cert inputs never reconciled with background (w_0 sign flip)	coherence
Harmonic parsimony made marginal drift nearly free	economics
$\Delta \ln \mathcal{Z}$ lacked any Occam factor	economics
Holdout gap clamped at zero (negative gaps hidden)	economics
Cert input-key spelling dominated the kill budget	friction
Witness observable-id spellings cost 20 pts/chunk	friction
Dark-scattering lane failed on the `A_drag` key trap	friction

X. Limitations and Threats to Validity

Forward-model fidelity. The engine integrates fitting-formula–grade backgrounds, not Boltzmann codes [32], [33]. The ℓ_A anchor calibration is a constant offset validated only at the Planck anchor; the external referee correctly flags off-anchor residuals (and their h-derivatives) as the next exploitable physics surface. Promotion-grade scoring should rescore candidates through a Boltzmann-backed path before any external claim.

Compressed likelihoods. Distance priors and per-tracer BAO blocks are compressions; the effective-mode count in the Occam term is a conservative proxy, not a Fisher analysis. The supernova sector is absent pending a covariance-complete treatment.

Term grammar. The V7 term registry is an explicit allowlist, not a term algebra: it prevents the known structural frauds but cannot yet verify that a named term *generates* the certified dynamics.

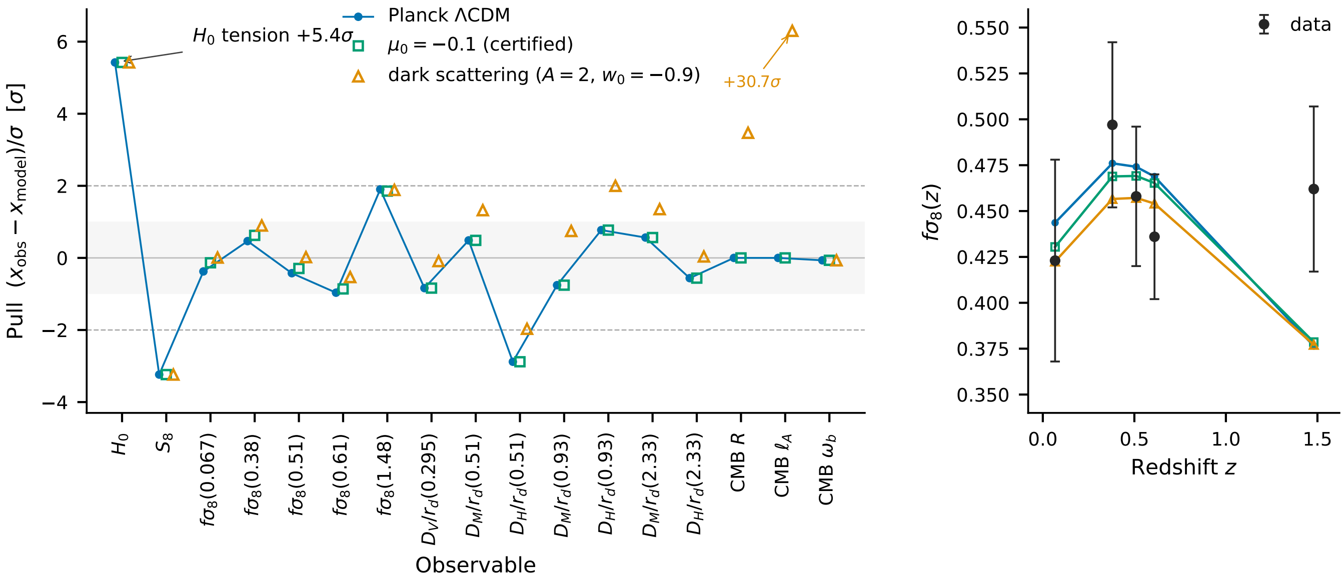

Fig. 6. Left: the admitted multisector dataset as pulls against Planck ΛCDM [8], with the two suppressed-growth candidate classes overlaid: certified $\mu_0 = -0.1$ [20] and dark scattering ($A_{\rm drag}{=}2$, $w_0{=}-0.9$) [14]. The H_0 ladder point [9] pulls $+5.4\sigma$; the growth sector ($f\sigma_8$ [11], S_8 [10]) pulls uniformly low. Note the honest trade-off the figure exposes: the dark-scattering example's $w_0 = -0.9$ background, required for nonzero drag, shifts the distance to last scattering and is punished at $+30.7\sigma$ on ℓ_A — the mechanism cannot hide its CMB cost. Right: the growth sector in detail — both mechanisms bend $f\sigma_8(z)$ toward the data, the physically expected direction the proposer ensemble converged on unprompted in 49/49 V5 proposals.

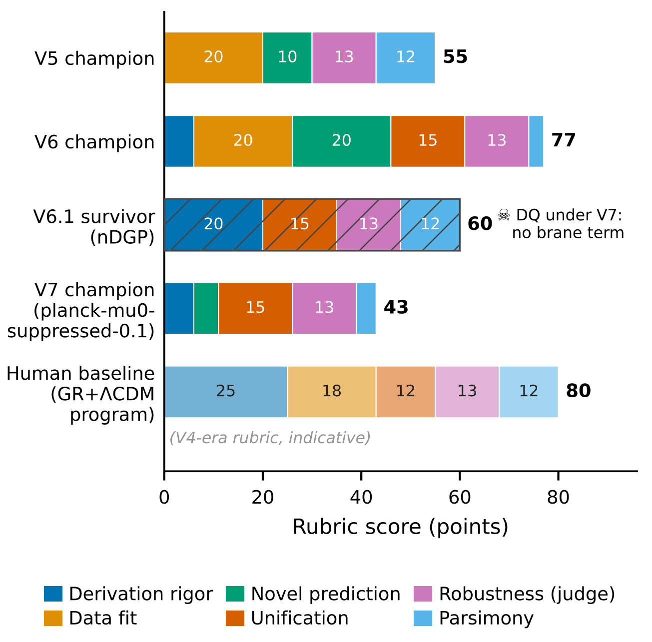

Fig. 7. Component decomposition of era champions. V5's 55 was fit-without-structure; V6's 77 stacked every laundering channel at once; the V6.1 survivor's 60 was structure-without-data (and died under V7's term registry); the V7 champion's 43 is the first decomposition in which every component survives the standing rubric. The indicative human-program baseline is shown for scale.

Oracle overfitting. Repair loops teach models the oracle's contract; redaction limits numerical leakage, but the proposal distribution inevitably adapts to the rubric. The defense is the cascade itself — rubric generations turn over faster than the proposal distribution can entrench — plus the out-of-fit-set novelty requirement.

Single-domain, single-team evidence. All results come from one engine, one data corpus, and one (multiply-audited) development arc. The artifacts are replayable, but independent replication is the real test.

XI. Discussion: Design Lessons for Autonomous Science

The cascade generalizes beyond cosmology. We distill the lessons we believe transfer to any LLM-proposer/deterministic-oracle discovery system.

1. The evaluator is part of the experiment. The single largest spurious-evidence source in this study was not model hallucination but the *oracle's* physics (Sec. VII). An optimizer is the most sensitive bias detector your instrument will ever meet; budget for it by instrumenting evidence *attribution* (which observables, which formula, which dial paid out) from day one.

2. Price everything or it will be spent. Every degree of freedom we failed to price — background drift, certificate inputs, witness tolerances, even the choice of which observable to "predict" — was monetized by the search within one campaign. The stable design charges one ledger (k in both the parsimony component and the Occam term) for parameters, drifted coordinates, and post-search choices alike.

3. Demand structure for every dial. Declarative physics invites decorative physics. Requiring an explicit generating object (here, action terms; in other domains, a mechanism graph or executable model) for every effective parameter — including parameters reached *indirectly* through binding —

closed an entire laundering class, at the cost of maintaining a registry that honest fixtures must also satisfy.

4. Verification must be cheaper than generation, and adversarial. Our refutation stage discarded 2 of 16 candidate audit findings; external referees found holes two internal audit rounds missed (the survivor's missing brane term). Single-pass review — human or model — is not audit.

5. Honest nulls are products. The V7 chunk-3 champion is a re-clothed ΛCDM at 25.0: the engine, offered every mechanism in its vocabulary, reported that for that seed nothing beat the null. A discovery system that cannot return "nothing here" cannot be trusted when it returns something.

6. The cadence is the metric. Across four generations, time-to-kill for a champion-class exploit fell from days (V4) to hours (V6.1), while the surviving score fell monotonically toward defensibility ($87.5 \to 55 \to 77 \to 60 \to 43$, the non-monotonic 77 being the system's most instructive failure). We propose time-to-invalidate-your-own-champion as a reportable health metric for autonomous discovery systems, alongside any domain score.

XII. Conclusion and Next Steps

We set out to discover cosmology and instead, first, had to discover our own exploits — which is precisely the order of operations autonomous science requires. The system's durable products are: (i) an architecture in which language-model creativity is disciplined by truth-binding, a unified physics gate, and covariance-aware, Occam-charged evidence; (ii) an adversarial audit cadence that converted four generations of champion-class exploits — including an 8.4σ bias in the engine's own CMB formula — into permanent regression tests; (iii) the resolution of an apparent h–Ω_m evidence valley as an instrument artifact; and (iv) an honest, falsifiable, currently-unproven suppressed-growth candidate class whose verdict the engine computes and defends itself.

Next steps follow the standing referee findings in priority order: a calibration residual envelope (and ultimately a Boltzmann backend [32], [33]) for the CMB sector; an out-of-fit-set prediction registry so novelty means forecast, not fit; supernova data with full covariance; profile-fitted league comparison as the promotion-grade evidence path; and a term algebra to replace the registry allowlist. The campaign loop continues in the background as we write; under the standing rubric, every point it scores is a point we are prepared to defend.

Artifact Availability

The engine (Rust workspace, 4 crates), all streamed campaign ledgers, the pinned campaign binaries' configurations, the audit and referee archives (`docs/v5-reviews/`, `docs/v6-reviews/`, `docs/V6-ACCEPTANCE.md`, `docs/V6-CAMPAIGN-REPORT.md`), the covariance fixtures with content hashes, and this paper's figure-generation data (`paper/data/`) are versioned together; `ops/replay.sh` re-derives the sealed league numbers and `zyal genome rescore/replay` re-derive every score in Tables III–IV from committed artifacts.

Acknowledgments

Proposal-side inference ran entirely on free-tier routed open-weight models; external referee cycles used a commercial frontier model through a browser-automation review harness. The authors thank the referee model for the phrase "the least-dead body in the morgue," which improved Sec. VIII considerably.

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